

CS641 Assignment 2

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Question 3

(a) Show that a cyclic group of n elements has $\phi(n)$ generators.

Let G be a cyclic group of order n , and let g be a generator of G . By definition, any element $h \in G$ can be written as $h = g^k$ for some integer k where $0 \leq k < n$, and $g^n = e$ where e is the identity element of G .

Let us find the conditions under which g^k is also a generator of G .
Let m be the order of g^k . By definition,

$$(g^k)^m = g^{km} = e.$$

From the properties of cyclic groups, $g^{km} = e$ if and only if (km) is a multiple of (n) . Let $d = \gcd(k, n)$. We can rewrite k and n as

$$k = d \cdot k', \quad n = d \cdot n',$$

where $\gcd(k', n') = 1$.

Using this, we get $(d \cdot k' \cdot m)$ must be a multiple of $(d \cdot n')$. Dividing both sides by d , we obtain $(k' \cdot m)$ must be a multiple of (n') .

Since $\gcd(k', n') = 1$, (m) must be a multiple of (n') .

We know that the order of an element is the smallest positive integer m satisfying the condition $(g^k)^m = e$, the smallest positive multiple of n' is $m = n'$.

Substituting $d = \gcd(k, n)$, we obtain

$$m = n' = \frac{n}{d} = \frac{n}{\gcd(k, n)}.$$

So the element g^k generates a subgroup of G whose order is given by:

$$|\langle g^k \rangle| = \frac{n}{\gcd(k, n)}$$

For g^k to generate the entire group G , its order must be equal to n . Therefore, we require:

$$\frac{n}{\gcd(k, n)} = n \implies \gcd(k, n) = 1$$

Thus, an element g^k is a generator of G if and only if k is coprime to n . The number of integers k in the range $0 \leq k < n$ that are coprime to n is defined by Euler's totient function, $\phi(n)$. As a result, a cyclic group of order n has exactly $\phi(n)$ generators.